Durham University

# Examination Paper

| Examination Session: | Year: | Exam Code: |
|---|---|---|
| May/June | 2020 | ENGI30120-WE01 |

**Title: MSc Electronic and Electrical Engineering**

## Electronics and Communications 3

| | | |
|---|---|---|
| Time Allowed: | 2 hours | |
| Additional Material provided: | None. | |
| Materials Permitted: | None. | |
| Calculators Permitted: | | Models Permitted:<br>Those from the Casio fx-83 and fx-85 series. |
| Visiting Students may use dictionaries: YES | | |

| | |
|---|---|
| **Instructions to Candidates:** | Answer ALL questions.<br>Use a separate answer book for each section.<br>All relevant workings must be shown. |

| Revision: | |
|---|---|

# Section A

## Question 1

(a) A discrete memoryless source produces outputs $x_1$, $x_2$, $x_3$, $x_4$, $x_5$ and $x_6$ with probabilities $\frac{1}{4}$, $\frac{1}{16}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{4}$ and $\frac{1}{16}$, respectively. Assume coding over an alphabet of size $D = 2$ (binary).

(i) Design a Shannon code for this source and determine its average codeword length.

(ii) Determine the entropy of this source. Comment on the result.

**[55%]**

(b) Consider a (15,11) Hamming code.

(i) What is the rate of this code?

(ii) Determine the minimum distance of this code.

(iii) How many errors can this code correct?

**[20%]**

(c) Consider the following amplitude-modulated signal

$$s(t) = [A_c + A_m \ \cos(8\pi 10^3 t)] \cos(18\pi 10^8 t).$$

(i) Determine the modulation index for $A_c = 5$ and $A_m = 4$.

(ii) Determine the power efficiency for $A_c = 5$ and $A_m = 4$.

(iii) Determine the values of $A_c$ and $A_m$ that maximise the power efficiency.

**[25%]**

**continued**

## Question 2

(a) Consider a discrete memoryless source that produces outputs $x_1$, $x_2$, $x_3$, $x_4$, $x_5$ and $x_6$ with probabilities $0.5$, $0.25$, $0.1$, $0.06$, $0.05$ and $0.04$, respectively.

(i) Design a Huffman code for this source over an alphabet of size $D = 3$ (ternary) and determine its average codeword length. Note that a ternary code is one which uses three alphabets in coding, e.g., $\{0, 1, 2\}$.

(ii) Determine the entropy of this source. Comment on the result.

**[55%]**

(b) Consider a wireless communication system with a band-limited additive white Gaussian noise (AWGN) channel with a bandwidth $W$, a transmit power $P$ and a noise with a power spectral density $N_0$.

(i) What is the capacity of this channel?

(ii) Explain what happens to the capacity of such a channel if we increase the transmit power $P$ without limit.

(iii) Explain what happens to the capacity of such a channel if we increase the bandwidth $W$ without limit, while $P$ and $N_0$ are kept constant. You can make use of $\ln(1 + x) \longrightarrow x$ as $x \longrightarrow 0$.

**[25%]**

(c) (i) Define frequency modulation (FM) and phase modulation (PM).

(ii) Draw a block diagram showing how to design a PM modulator using an FM modulator.

**[20%]**

**continued**

# Section B

## Question 3

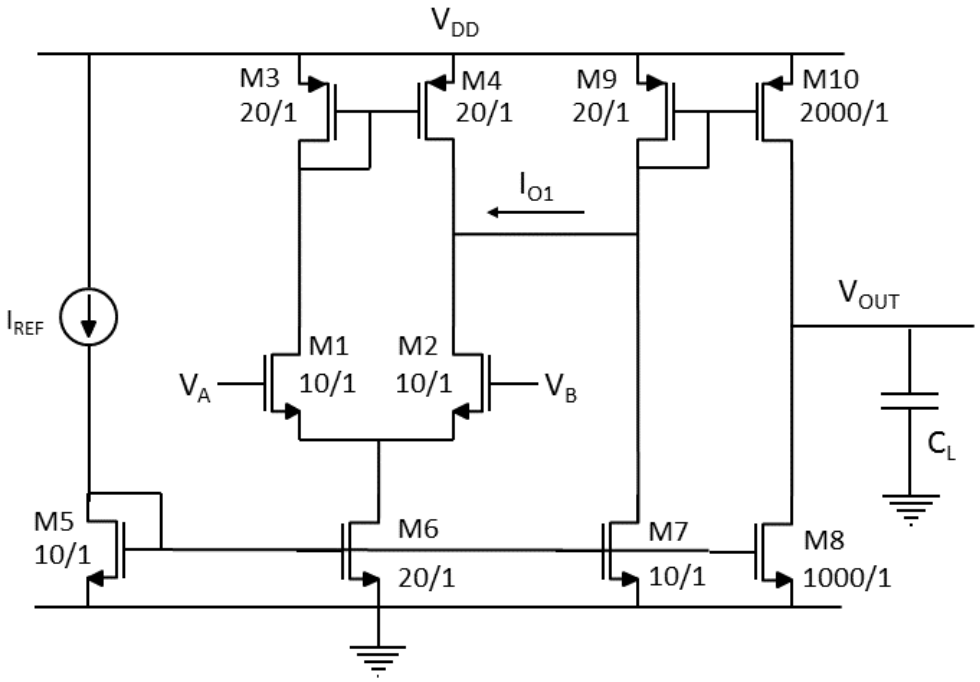



Figure Q3: The differential amplifier.

The circuit of a differential amplifier is shown in Figure Q3. The MOSFET parameters and the values of other circuit elements are:

$V_{DD} = 3$ V

$I_{REF} = 30$ $\mu$A

$\mu_n C_{OX} = 150$ $\mu$A/V$^2$

$\mu_p C_{OX} = 50$ $\mu$A/V$^2$

$V_T = 0.5$ V

$C_L = 200$ pF

$V_M = 10$ V

The aspect ratios W/L of the MOSFETs are shown in the circuit.

(a) Explain how the circuit operates and derive an expression for the differential DC gain, clearly showing the dependence on the reference current $I_{REF}$ and indicate which of the input terminals is the positive or inverting. *Hint:* Consider first the dependence of the current $I_{O1}$ on the differential input and its effect on the current mirror.

**[50%]**

**continued**

(b) (i) Derive an expression for the frequency response as a function of the current $I_{REF}$, neglecting the parasitic capacitors of the MOSFETs, and state the dependence of the gain-bandwidth product on $I_{REF}$.

(ii) Sketch the Bode plots (magnitude and phase) of the frequency response, clearly stating the value of key parameters.

**[50%]**

**continued**

## Question 4

(a) Explain how a CMOS inverter operates and state its main advantages. Hence sketch the transfer characteristic of a CMOS inverter and state the operating regime of each MOSFET along the output/input characteristic.

**[30%]**

(b) We want to design a CMOS half-adder. The truth table of the half-adder is:

| A | B | SUM | CARRY |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

where A and B are the inputs and SUM and CARRY are the binary sum and the output carry, respectively.

(i) Draw two symmetric CMOS circuits which realise the SUM and CARRY.

(ii) Determine the channel aspect ratio of all MOSFETs, assuming that the smallest acceptable aspect ratio of an n-MOSFET is $(W/L)_N = 8$ and that $\mu_n/\mu_p = 4$.

**[50%]**

(c) (i) Define the dynamic power dissipation of a logic gate.

(ii) Assuming that $V_{DD} = 3$ V and that the gate capacitance of a MOSFET with a channel aspect ratio W/L is $C_G = 10\times$(W/L) fF, find the power dissipated by the half-adder when B $= 1$, and A is clocked at a frequency $f_{CLK} = 1$ GHz. Neglect the effect of the load capacitance on SUM and CARRY.

**[20%]**

**END**